

JONATHAN WOO

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RESEARCH INTERESTS

Deep learning, drug discovery, protein engineering, AI for science, geometric deep learning, quantum mechanical modelling

EDUCATION

University of Toronto

Bachelor of Applied Science in Engineering Science, Major in Machine Intelligence
5x Dean's Honour List

Sept. 2020 - Apr. 2025

cGPA: 3.67/4.0

Major GPA: 3.95/4.0

RELEVANT COURSEWORK

• ECE367: Matrix Algebra & Optimization	(A)	• ECE353: Systems Software	(A+)
• ECE421: Introduction to Machine Learning	(A+)	• ECE368: Probabilistic Reasoning	(A+)
• ECE358: Foundations of Computing	(A)	• ROB311: Artificial Intelligence	(A+)
• ECE324: Software & Neural Networks	(A+)	• STA410: Statistical Computation	(A+)

GRANTS & AWARDS

Hosinec Family Scholarships

Mar. 2023

University of Toronto

- Academic merit, leadership, and financial need.

Shaikh And Hashemi Family Engineering Science Award

Oct. 2022

University of Toronto

- Academic merit, leadership, and financial need.

Hok Chee Poon & Yim Hung Kwong Bursary

Dec. 2021

University of Toronto

- Academic merit, leadership, and financial need.

Fu Siang Pang & Ying Au Yeung Bursary

Dec. 2021

University of Toronto

- Academic merit, leadership, and financial need.

Class of 4t7 Scholarship

Dec. 2021

University of Toronto

- Academic merit, leadership, and financial need.

Class of 2004 Grant

Dec. 2021

University of Toronto

- Academic merit, leadership, and financial need.

Faculty of Applied Science and Engineering Admission Scholarship

Oct. 2020

University of Toronto

Army, Navy and Air Force Veterans in Canada (ANAVETS) Cadet Medal of Merit

Aug. 2018

Air Cadet League of Canada

- The ANAVETS Cadet Medal of Merit is awarded by the Army, Navy, and Air Force Veterans in Canada to recognize excellence among cadets in the Canadian Cadet Organization. One medal is presented at each Cadet Training Centre (CTC) to the top male and female cadets in specified courses. Recipients are selected based on overall achievement, leadership, instructional abilities, dress, deportment, motivation, and teamwork.

RESEARCH EXPERIENCE

Undergraduate Thesis Student

Toronto, Canada

Vector Institute — [Chemspacelab](#)

Sept. 2024 - Apr. 2025

Supervisor(s): Anatole Von Lilienfeld

- Developed machine learning and symbolic regression models to optimize parameters for semi-empirical tight binding (xTB), achieving near-DFT accuracy while reducing computational complexity from $\mathcal{O}(n^4)$ to $\mathcal{O}(n^2)$.

Visiting Research Student

Cambridge, USA

MIT, Harvard Medical School — [Laboratory for Translational Engineering](#)

Sept. 2023 - Aug. 2024

Supervisor(s): Alvin Chan, Troy Kang, Giovanni Traverso

- Developed COMET, an SE(3)-Transformer for mRNA LNP drug discovery. Leveraged multitask contrastive learning, noise augmentation, and model ensembling to enhance prediction accuracy and robustness, significantly accelerating candidate identification.

- Led successful lead generation and optimization of novel formulations, achieving cell and organ selectivity, cryoprotection, lyoprotection, and thermostability in vitro and in vivo. Patented formulations with >100× efficacy than clinical baselines.
- Employed integrated gradients and analysis of latent embeddings to derive mechanistic insights into LNP design, enhancing the understanding of structure-function relationships. Uncovered novel insights to factors contributing to efficacy.
- Published LANCE, the largest LNP dataset, spanning extensive chemical space, a key resource for the research community.
- Contributed as a core team member to [ARPA-H REO's](#) \$70M+ initiative, advancing metabolic disease treatment research.
- Built an innovative time series-based machine learning algorithm with a custom attention mechanism to balance impedance and optical sensors, demonstrating the capability of algorithmically characterizing embryonic stem cells.
- Pioneered an in-context learning paradigm for scientific discovery, enabling large language models (LLMs) to leverage scientific foundation models without the need for fine-tuning. This approach involved injecting foundation model embeddings directly into the LLM's text embedding space, enabling inference conditioned on both the training corpus and external model embeddings, starting with molecular property prediction.
- Developed an optimized pipeline and user-friendly interface for LLM inference, fine-tuning, in-context learning, and evaluation, leveraging multi-GPU parallelism across distributed clusters to drastically improve the research team's efficiency and scalability.

Computational Neuroscience Research Intern

University of Toronto — [CoNSens Laboratory](#)

Supervisor(s): Matthias Niemeier

Remote

May - Dec. 2023

- Developed a tool to visualize the activity of convolutional layers in CNNs, enabling the neuroscience team to analyze complex, long-range interactions.

Machine Learning Research Intern

Canadian Nuclear Laboratories — Applied Physics Branch

Supervisor(s): Jude Alexander, Oleg Kamaev

Chalk River, Canada

May - Aug. 2022

- Developed machine learning models to classify nuclear material using low resolution time-series gamma ray spectra. Attained a state-of-the-art 94% classification accuracy using a random forest classifier, effectively minimizing nuisance false alarms.
- Implemented training and inference parallelization with CUDA and HPC which reduced the average processing time by 80%, thereby accelerating testing and optimization.

AI Hardware Acceleration Research Intern

University of Toronto — [Intelligent Sensory Microsystems Laboratory](#)

Supervisor(s): Amiralí Amirsoleimani, Roman Genov

Toronto, Canada

Apr. 2021 - Apr. 2022

- Formulated 4 novel algorithms and 3 metrics for memristor crossbar arrays (MCAs) - analog hardware deep learning accelerators enabling $\mathcal{O}(1)$ complexity of matrix-vector-multiplication - to predict device behaviour using physical properties such that manufacturers can alleviate the bottleneck of testing individual devices.

INDUSTRY EXPERIENCE

Software Engineer Intern

The Trade Desk

Hong Kong

Jun. - Aug. 2023

- Spearheaded the development of a client onboarding tool, alleviating a critical bottleneck and retiring an inefficient scripting process. This saved engineers on average 10 hours a week and reduced the total onboarding time by 3 weeks.
- Led the retrofitting of two microservices to GraphQL, maintained gRPC compatibility, and reduced 3-5 API calls into one.
- Refactored 5+ critical endpoints, unit tests, and end-to-end tests improving performance and long-term maintainability.

Machine Learning Developer

Omni Research — [beta.omnilabs.ai](#)

Toronto, Canada

Sept. 2022 - Apr. 2023

- Developed a ChatGPT-like interface and crafted prompts for GPT-3 retrieval-augmented generation (RAG), enabling a personalized chatbot with persistent browsing capabilities.
- Created 7 few-shot learning prompts and APIs to query GPT-3, enabling the automatic generation of explanations, counterarguments, and Tweets. Directly expanded the capabilities of the platform with 15% more users post-launch.

JOURNAL PUBLICATIONS

RNA Nanomedicine Stabilization with Deep Learning

Science Advances

Jonathan W*, Alvin C.*, Ameya R. K.*, et al. Giovanni Traverso

Under Review

Lipid Nanoparticle Design with Composite MatErial Transformer (COMET)

Nature Nanotechnology

Alvin C.*, Ameya R. K.*, Qing R. Q., Xisha H., **Jonathan W.**, et al. Giovanni Traverso

Under Review

Smart Needle System for Point-of-care Characterization of Cell Therapy

Nature Biotechnology

Yiyuan Y.*, Ziliang K.*, Nabil S.*, Jong S. L.*, Ming Z.*, Fangzhou X., Injoo M., **Jonathan W.**, et al. Robert Langer, Giovanni Traverso

Under Review

Oesophageal Tissue Screening (OTiS) Platform

Under Review

Nature Biomedical Engineering

Christina K.*, Thomas Y., Alvin C., Nikhil L., Charmaine J. C., Shriya R., Yuebin H., **Jonathan W.**, et al. Giovanni Traverso

CONFERENCE PUBLICATIONS

LLMxFM: Fine-Tuning Free Synergy of Foundation Models & Large Language Models

In Progress

2025 International Conference on Machine Learning (ICML)

Jonathan W.*, Alvin C.*, Pari L., Deepak A. S., Giovanni Traverso

Modelling Tools for Enhancing Nuclear Border Security

May 2024

International Conference on Nuclear Security: Shaping the Future

Genevieve H.*, **Jonathan W.**, et al.

Radioisotope Identification Using CLYC Detectors

Dec 2022

2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)

[10.1109/ICMLA555696.2022.00214](https://doi.org/10.1109/ICMLA555696.2022.00214)

David P. L.*, Jude A.*, (Acknowledgements) **Jonathan W.**, et al.

PROJECTS

LLMs for Addiction Recovery & Therapy

Sept. - Dec. 2024

[OpenRecovery](#)

- Collaborated with a startup to transition model deployment from prompt-engineered in-context learning to fine-tuned large language models (LLMs) optimized via Direct Preference Optimization (DPO) using LoRA.
- Designed and implemented a scalable, automated evaluation pipeline utilizing LLM-as-a-Judge to assess approximately 1,000 conversations, creating a high-quality dataset of preference pairs for improved model training and evaluation.

Decision Transformer for Autonomous Highway Driving

Jan. - Apr. 2023

github.com/caixunshiren/Highway-Decision-Transformer

- Developed an autonomous highway driving agent using Decision Transformers, outperforming top RL algorithms (Deep Q-Learning, PPO) with superior adaptability and decision-making in complex scenarios.
- Created simulation environment and scripts to collect 20000 episodes of training data. Formulated novel reward-functions to influence driving behaviours from aggressive to passive.

IMC Prosperity Global Trading Challenge - #7 Canada/Top 1% Worldwide

Apr. 2024

github.com/Jonathan-Woo/prosperity

- Led the end-to-end software implementation of quantitative trading strategies, building and maintaining essential tools such as an order book, backtester, and real-time performance dashboards. Enhanced trading infrastructure by integrating low-latency data feeds, optimizing execution algorithms, and ensuring robust data integrity for accurate trade simulations and live trading.
- Utilized historical price data to rigorously backtest and fine-tune parameters across a suite of models, including ARIMA for time-series forecasting, pairs trading for correlated asset strategies, mean reversion for statistical arbitrage, and linear regression for predictive pricing models. Performed parameter optimization and sensitivity analysis to refine model accuracy and improve profitability across market regimes, aligning with risk management objectives.

SKILLS

Languages: English (native)

Programming Languages: Python, C, C++, C#, TypeScript, SQL, MATLAB, $\text{\LaTeX}$

ML: TensorFlow, PyTorch, JAX, Scikit-learn, Hugging Face, LangChain, XGBoost, RDKit, PyMC, Pandas, NumPy, SciPy

Backend: Flask, Docker, Kubernetes, gRPC, REST, GraphQL

HPC: CUDA, SLURM